

AGRICULTURAL ROBOT S.A.S.

PRESENTED BY

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APPLIED TECHNOLOGIES

TECHNOLOGY PROGRAMS

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Introduction

An automated agricultural robot is a device designed to carry out highly specific agricultural tasks without the need for constant human intervention. These robots are equipped with components such as sensors, actuators, and advanced software that enable them to perform a wide variety of agricultural activities, including planting, harvesting, crop spraying, and monitoring environmental conditions, among others.

This project is mainly focused on the Colombian agricultural sector, specifically on small-scale farmers in the Cundinamarca region, with the purpose of optimizing and promoting community development through the inclusion of new technologies and processes that will help facilitate crop maintenance using more reliable and lower-cost methods.

Background

The transition toward agriculture possibly represents one of the most transformative processes in the history of humanity. This paradigm shift not only reconfigured subsistence systems and the organization of human settlements, but also profoundly altered ecosystems and territorial organization. Likewise, it established the material conditions necessary for the development of complex social structures, the emergence of social stratification, and ultimately, the civilizational processes that define contemporary social organization.

Similarly, it is important to argue that without the rise of agriculture, the demographic viability of our species would have been severely limited, restricting its global expansion or even compromising its very survival. Although advances in scientific methodologies applied to the analysis of archaeological remains have significantly increased our knowledge about the origins of agricultural practices, current historiography remains fragmented in nature. In many cases, the dominant narrative has been constructed from an androcentric perspective, which at times lacks a profound epistemological connection with the pragmatic, everyday, and cumulative dimensions of agricultural labor.

Likewise, these narratives often overlook the technical complexity inherent in agricultural practices, including the systematic selection of genetic material (plants, seeds, and animals), adaptation to climate variability, strategies for managing production cycles, and the comprehensive use of agricultural biodiversity. Consequently, conventional historiographical narratives reveal a noticeable disconnection from the current practices of peasant and Indigenous communities.

Therefore, the present text not only incorporates the available scientific evidence, but also proposes an articulation between historiography and traditional knowledge systems, with a particular emphasis on the knowledge generated by women in rural contexts. This approach is essential for understanding the ethical and material dimensions of the caregiving practices that have historically ensured the continuity of agriculture.

Justification

The purpose of this project is fundamentally based on increasing the efficiency and productivity of small-scale farmers in the Cundinamarca region during the planting and monitoring of crops. This will result in more abundant and higher-quality production without the need for constant human intervention. The inclusion of agricultural robots provides local farmers with more direct access to advanced technologies that could otherwise remain out of reach due to various factors. This initiative will encourage the modernization and future competitiveness of Colombian agriculture in the global market.

This project also demonstrates a contribution to the sustainable development of rural communities in Cundinamarca, while reflecting innovation in modern agriculture. In addition, it promotes training and education in new agricultural technologies to address the different challenges faced in today's agricultural sector.

Theoretical Framework

Precision Agriculture

Precision agriculture is a system used to analyze and control the spatiotemporal variation of land and crops. Spatial variation refers to the differences in soil fertility and plant distribution across different plots of land where implementation takes place. Temporal variation involves differences in the productivity of the same land from one season to another.

Stages of Precision Agriculture

Data Collection:

This stage is carried out using cameras, specialized equipment, and sensors to measure the dimensions and characteristics of the land.

Data Analysis:

An expert analyzes the collected data in order to properly manage the spatiotemporal variation detected during the data collection process.

Implementation

Guided by precision agriculture (PA), the producer cultivates the land according to the provided specifications.

Ultrasonic or Proximity Sensors (LIDAR)

These sensors are used to measure the distance between the planting device and the ground, ensuring that seeds are placed at the appropriate depth.

Actuators

Actuators are devices that transform electrical energy into mechanical movement. These may include electric motors, servo motors, pumps, and valves that are used to control the direction and speed of the robot, as well as to perform specific tasks in the field.

Legal Framework

1. Framework for Agricultural Supplies and Equipment

Law 2183 of 2022:

This law is fundamental because it establishes the National System for Agricultural Inputs in Colombia. It defines and promotes the use of small-scale machinery for land preparation and planting activities, providing legal support for the adoption of automation technologies aimed at improving rural productivity.

2. Technical Safety Standards (Machinery)

Although the design is autonomous, it must comply with industrial safety standards in order to prevent accidents in the field.

Operational Safety:

It is necessary to consider principles similar to those established for traditional agricultural machinery (such as tractors), including protective guards on moving parts, emergency stop systems (panic buttons), and visual or sound warning signals.

International Standards (ISO 25119):

Although these are voluntary technical standards, it is recommended that the development of the electronic control system and software be aligned with the ISO 25119 series (Tractors and machinery for agriculture and forestry – Safety-related parts of control systems). This ensures that, in the event of software or hardware failure, the robot behaves safely, for example through a complete shutdown or automatic stop.

3. Data Protection and Privacy

Since the robot uses sensors (cameras and GPS) to map land or collect crop information, it must implement measures related to data protection.

Law 1581 of 2012 (Data Protection Law):

Considering that the robot may collect data associated with individuals (such as images of workers or georeferenced information from private properties), the project must guarantee the proper handling of this information, prevent unauthorized access, and comply with the privacy policies established by the Superintendence of Industry and Commerce.

4. Navigation and Autonomy

Use of Frequencies:

Since the robot communicates through radio frequencies (especially when using high-precision GPS or telemetry systems), it must comply with the regulations established by the Ministry of Information and Communication Technologies (MinTIC) regarding the radioelectric spectrum in order to avoid interference.

Research Problem

The agricultural sector has historically been one of the fundamental pillars for the economic and social development of human societies. Since the earliest agricultural settlements, food production has been directly linked to population growth, becoming a determining factor for the stability and progress of civilizations. In the modern era, with the rapid increase of the world population, the need for greater food production has become a major concern for governments and international organizations.

In the global context, the 2008 World Development Report, entitled *Agriculture for Development*, stated that agricultural production would need to increase by 50% in order to supply the world population in the coming decades (Washington, 2007). This call to action was based on the reality that food demand has grown at a faster rate than the productive capacity of many agricultural sectors. Population growth and urbanization, together with the challenges of climate change, have placed pressure on agricultural production, requiring new methodologies and technologies that optimize resources and increase productivity.

In the case of Colombia, the department of Cundinamarca has played a key role in the country's agricultural production, with a territorial extension of approximately 24,000 km², making it one of the main productive regions of the Cundiboyacense highlands. According to data from Terridata, the region has a high percentage of potato cultivation (77.31%), along with carrot and tomato crops. These crops form part of the agricultural foundation of the region, which is characterized by a cold and temperate climate ideal for certain agricultural products, but also vulnerable to climatic phenomena such as frost, as well as El Niño and La Niña events. (Cristian Castro Santana, Iván Carrillo Rodríguez, 2022)

Historically, the economy of Cundinamarca has been centered on agriculture and livestock farming, although in recent decades the extraction of natural resources has also gained importance. However, one of the persistent challenges in the region is the low adoption of modern technologies in the agricultural sector, which has limited growth potential and resource optimization. Despite the fact that Cundinamarca has one of the largest rural areas in the country (more than 35% of the territory), the use of advanced technologies, such as precision agriculture, has been limited. This has resulted in significant crop losses due to the lack of proper control and monitoring, as well as vulnerability to climate change. (Cristian Castro Santana, Iván Carrillo Rodríguez, 2022)

Research Question

How could the processes and methods involved in the planting of fine-grain seeds for the production of white wheat flour be optimized among small-scale farmers in Cundinamarca?

General Objective

To build an autonomous vehicle that optimizes conventional, artisanal, and traditional methods used in the planting of corn seeds.

Specific Objectives

To design the layout and structural plan of the components of an agricultural robot.

To research different electronic and mechanical devices that facilitate and adapt to the robot's needs based on minimum technical requirements.

To implement an agricultural robot with appropriate technology that improves and optimizes planting processes in the agricultural sector.

To optimize the use of resources such as seeds and water utilized for crop cultivation.

methodology

Aplicación metodología al proyecto

01.

PLANIFICACION

Definir objetivos, identificar alternativas y restricciones



03.

DESARROLLO Y PRUEBA

Diseñar, codificar, implementar y probar.



02.

ANALISIS DE RIESGO

Evaluar riesgos, seleccionar el enfoque adecuado, prototipar si es necesario.



04.

EVALUACION

Obtener retroalimentación de los usuarios y hacer ajustes.



Modelo de Espiral

Metodología



Development of the Research Methodology

a. Cycle Planning

1. Objectives

To build an autonomous vehicle that optimizes conventional, artisanal, and traditional methods used in the planting of corn seeds.

Alternatives

Different methodologies or approaches for the research will be analyzed (quantitative, qualitative, or mixed methods). The approach that best fits the research questions will be selected.

Constraints

The general constraints of the project include time, cost, and scope. Understanding these limitations is important because they directly affect the project's outcomes.

Time: 2 and a half years

During the current semester, our main priority is focused on developing a preliminary design of the project. This strategy is based on the understanding that, as the academic cycles progress, we will acquire the essential knowledge required to support the advancement of the prototype's development. However, it is important to emphasize that future progress will be closely linked to the objectives previously established by the work team.

Cost: \$500,000 COP

Scope:

Throughout this semester, our primary goal is to acquire the necessary knowledge to pre-design the prototype of our autonomous robot. Based on this, we aim to achieve progressive development throughout future semesters, as we gather more information and further define the actual scope of the project.

b. Risk Analysis

Risk Identification

Possible challenges that may arise during the research process include:

1. Lack of access to data.
2. Problems in data collection.
3. Financial limitations for funding the project.
4. Established time constraints.
5. Limited access to certain materials or components required for the construction of the robot.

Risk Mitigation

Strategies will be designed to minimize and manage these risks effectively.

c. Development and Testing

Plan Execution

1. Prototype designs.
2. Piece-by-piece design development.
3. Identification of sensor applications.
4. Planning the placement of each sensor.
5. Collection of remaining materials needed for the construction of the prototype.
6. Simulations using the Tinkercad platform.

Testing

Preliminary research results will be analyzed. For example, if an experiment is conducted, results will be reviewed to determine whether they are consistent and whether variables are properly controlled. (To be developed in the future.)

d. Evaluation

Feedback

The obtained results will be reviewed, and the research will be adjusted according to the findings. This may require refining the methodology, adjusting the approach, or even modifying the research question.

Plan for the Next Cycle

Once the evaluation is completed, the next cycle will be planned. This may involve another phase such as data analysis or drafting the research report.

Cycle 1: Theoretical Framework and Research Questions

Planning:

Define the objectives and research questions.

Risk Analysis:

Evaluate whether the questions are viable and whether sufficient literature exists.

Development:

Conduct bibliographic research and define the theoretical framework.

Evaluation:

Review the findings and adjust the questions if necessary.

Cycle 2: Research Method Design

Planning:

Select the research method (quantitative or qualitative).

Risk Analysis:

Evaluate issues such as access to data, validity, and reliability.

Development:

Design the methodology (surveys, interviews, experiments).

Evaluation:

Validate whether the method is appropriate through pilot testing or expert feedback.

Cycle 3: Data Collection and Analysis

Planning:

Collect the necessary data.

Risk Analysis:

Consider issues related to the sample or errors in data collection.

Development:

Conduct experiments, surveys, or data collection activities.

Evaluation:

Carry out an initial analysis of the data to validate consistency and quality.

Cycle 4: Interpretation and Conclusions

Planning:

Interpret the results.

Risk Analysis:

Ensure that the conclusions are well supported.

Development:

Write the conclusions.

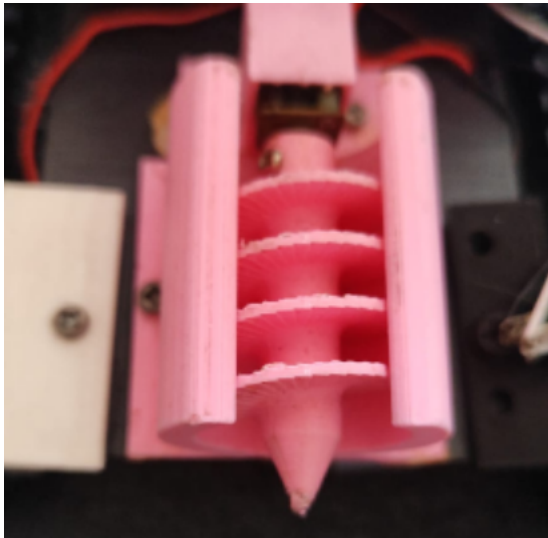
Evaluation:

Submit the work for review by experts or academic committees and make adjustments to the final report if necessary.

Cronograma de actividades											
Objetivo específico	Nombre de la actividad -	Duración de la actividad (mes)									
		octubre 1	Noviembre 2	diciembre 3	enero 4	febrero 5	Marzo 6	abril 7	Mayo 8	juno 9	julio 10
Objetivo específico 1.	Actv. 1 Investigación Y planificación										
	Actv. 2 Analisis de riesgos y resultados										
	Actv. 3 Desarrollo y implementacion										
Objetivo específico 2.	Actv.1 Investigación Y planificación										
	Actv. 2 Analisis de riesgos y resultados										
	Actv. 3 Desarrollo y implementacion										
Objetivo específico 3.	Actv.1 Investigación Y planificación										
	Actv. 2 Analisis de riesgos y resultados										
	Actv. 3 Desarrollo y implementacion										

Robot Components

Endless Screw



Operating Principle

When the motor rotates, the screw (the pink helical component) creates a loading chamber between each of its blades or spirals.

Loading:

The corn seeds fall by gravity from a hopper (which should be located at the upper part) and are deposited into the spaces between the spirals.

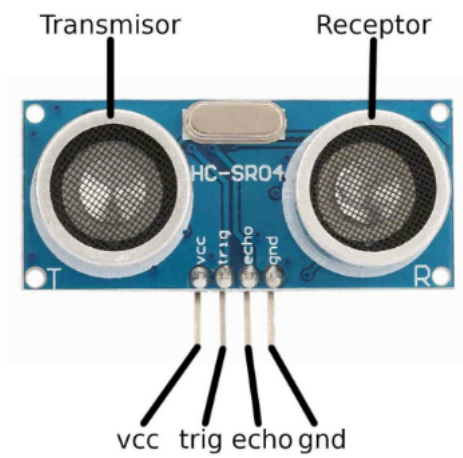
Transport:

As the screw rotates inside the casing (the cylindrical body surrounding it), the seeds are pushed in a controlled manner toward the tip.

Discharge:

The geometry of the conical tip helps direct the flow of seeds toward the exact planting point, preventing blockages or jams.

Ultrasonic Sensor



Specifications

Supply Voltage (VCC): 5V DC.

Current Consumption: Approximately 15 mA.

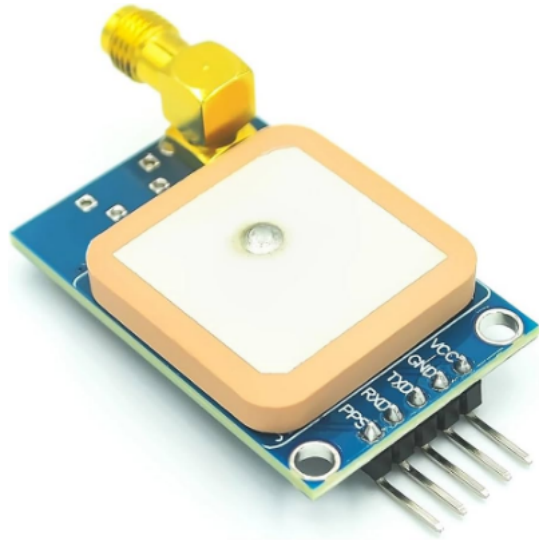
Measurement Range: From 2 cm up to 400 cm (4 meters).

Detection Angle: Approximately 15 degrees (the sound beam is conical rather than a laser point).

Accuracy: Approximately 3 mm.

Operating Frequency: 40 kHz.

GPS NEO-7M



Specifications

Operating Voltage: 2.7V to 5.0V (typically powered with 3.3V or 5V).

Current Consumption: Approximately 35 mA.

Communication Interface: UART (TTL levels, compatible with 3.3V / 5V systems).

Default Baud Rate: 9600 bps.

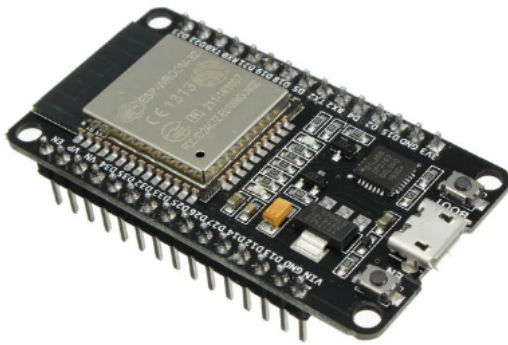
Operating Frequency: L1 band (1575.42 MHz).

GNSS Capability: Supports GPS, GLONASS, QZSS, and SBAS.

EEPROM Memory: Many modules include EEPROM memory to store parameter configurations, allowing the device to retain settings even after power loss.

Startup: Supports hot starts thanks to the rechargeable backup battery integrated into the module.

Microcontroller



Processor and Memory

Microcontroller: Tensilica Xtensa Dual-Core 32-bit LX6.

Clock Speed: Adjustable from 80 MHz up to 240 MHz.

Flash Memory: Generally 4 MB (used to store the program code).

SRAM Memory: 520 KB (used for variables and real-time process management).

Wireless Connectivity (Key for the Robot)

Wi-Fi: IEEE 802.11 b/g/n standard (up to 150 Mbps). It can operate as an access point or connect to an existing network.

Bluetooth: Dual mode (Classic Bluetooth and Bluetooth Low Energy - BLE). Ideal for controlling the robot through a mobile application.

Inputs and Outputs (GPIOs)

Digital Pins: Includes a large number of configurable pins.

Analog-to-Digital Converter (ADC): Features 12-bit resolution, allowing more accurate readings from analog sensors, such as soil moisture sensors, compared to an Arduino Uno.

PWM: Capable of controlling the speed of the robot's motors (requires H-bridge drivers such as the L298N or TB6612FNG for motor operation).

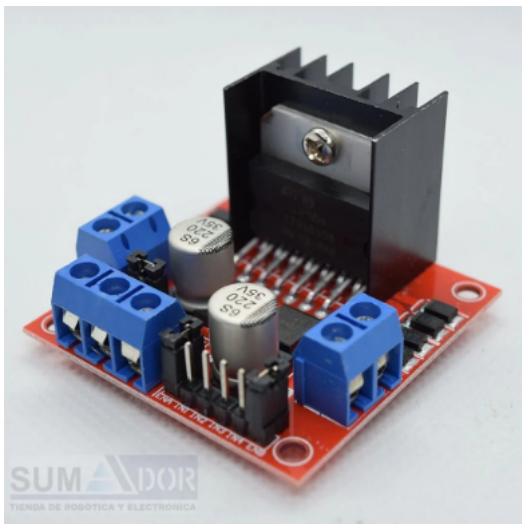
Communication Protocols: Supports UART (for the GPS module), I2C (for displays or advanced sensors), and SPI.

Power Supply and Energy

Input Voltage (VIN): Can be powered through the micro-USB port (5V) or through the VIN pin (recommended between 5V and 12V).

Operating Voltage: 3.3V.

H-Bridge:



Specifications

Motor Operating Voltage (VMS/VIN):

Capable of handling DC motors from 5V to 35V.

Maximum Current:

Up to 2A per channel (it can independently control two motors simultaneously).

Module Power Consumption:

Approximately 2 mA.

Maximum Power:

25W.

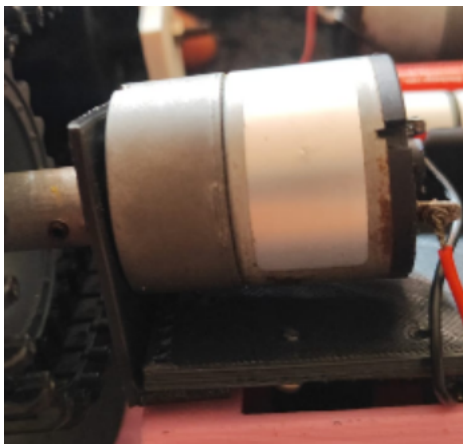
Control Logic Levels:

Operates with 5V TTL signals. Although the ESP32 works at 3.3V, the L298N usually recognizes 3.3V signals as a high logic level (“HIGH”). However, if the motors do not start with enough power, a logic level converter may be required.

Thermal Protection:

Includes a heat sink (the large black block on top) because the chip heats up when handling high power levels.

Tracked Motor



Specifications

Operating Voltage:

Generally designed to operate between 6V and 12V DC.

Speed (RPM):

These motors usually include gear reductions that provide between 60 and 300 RPM at the output shaft. For an agricultural robot, lower RPM motors are preferred because they provide greater torque and strength to move a loaded structure.

Torque (Force):

Thanks to the gearbox (the metallic component connecting the motor to the wheel), these motors deliver high torque, allowing the robot to overcome slopes and move across uneven terrain while carrying weight.

Current Consumption:

No-load: Approximately 200 mA – 400 mA.

Under load (stalled or forced movement): Can increase up to 2A or more.